INCLUSIFY

AI-Powered LGBTQ+ Inclusive Language Analyzer for Academic Texts

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View on GitHub

Try Inclusify

The Purpose of this system:

The Problem

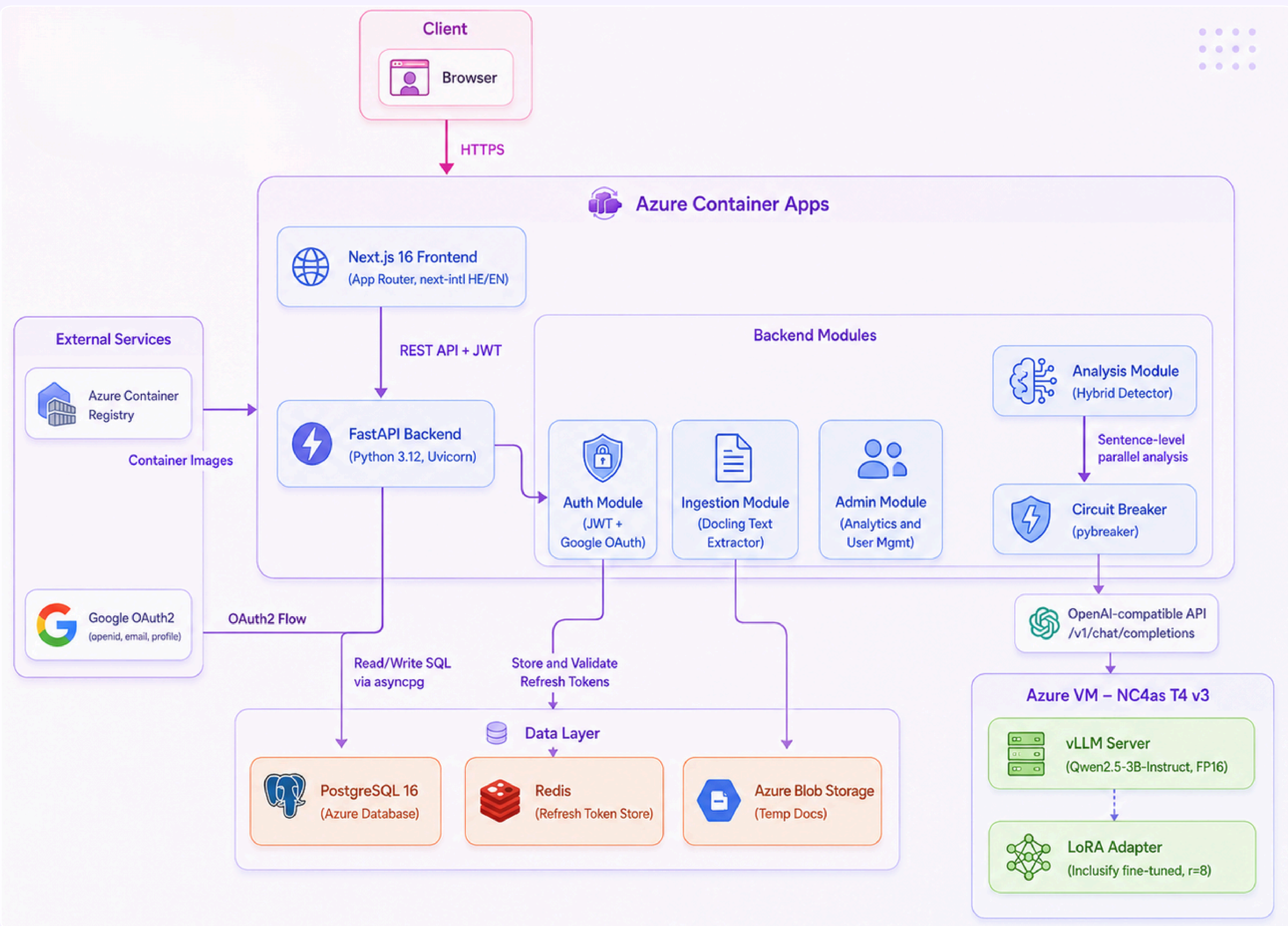
Academic texts may unintentionally include outdated, biased or exclusionary language regarding sexual orientation and gender identity. Existing writing tools usually focus on grammar and style, but do not provide specialized LGBTQ+ inclusive-language guidance.

Our Solution

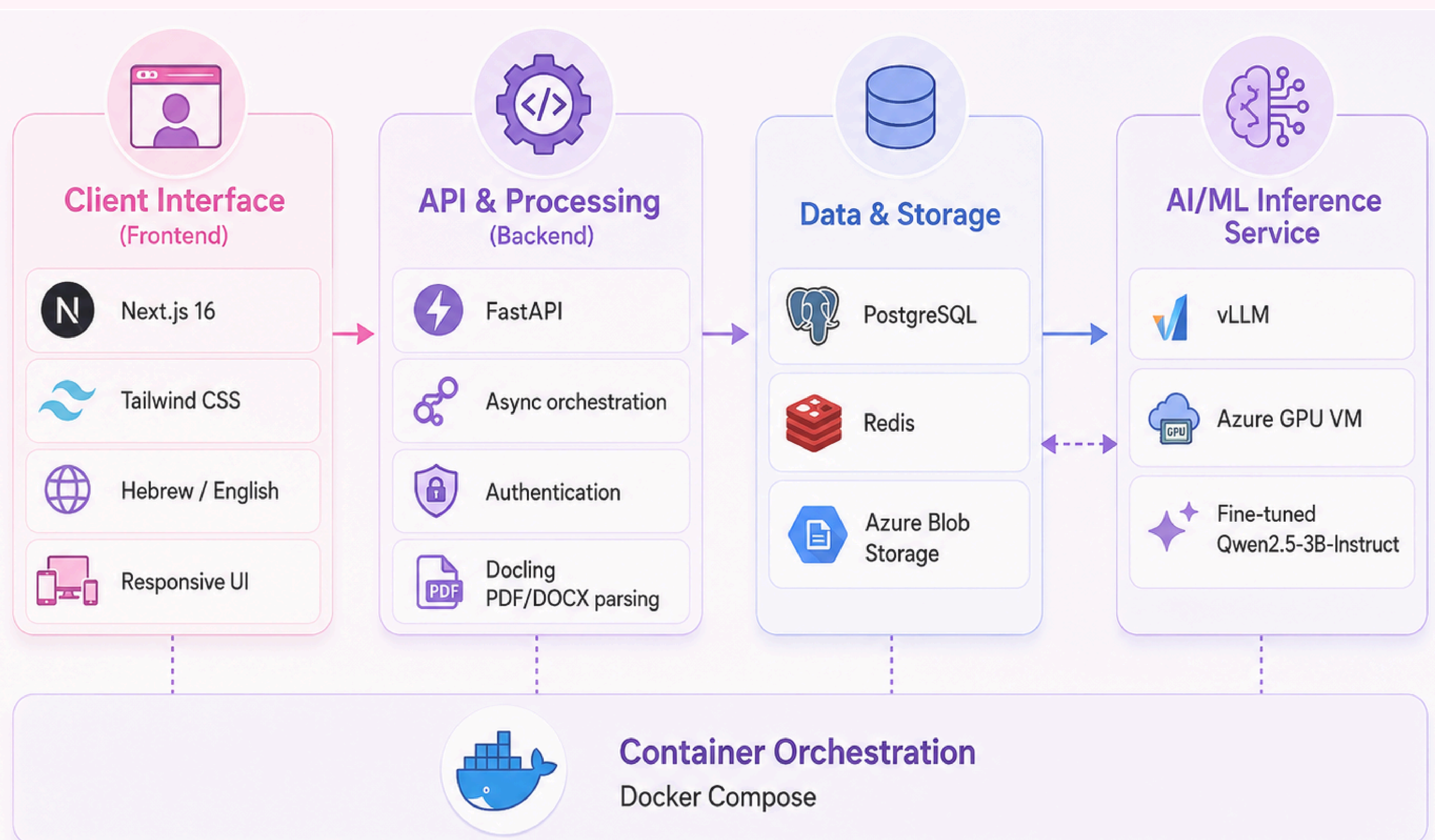
Inclusify is a bilingual web platform that analyzes academic texts in Hebrew and English. It identifies potentially problematic language, highlights the relevant phrases, explains why they may be inappropriate and suggests more inclusive alternatives.

Target Users: Students, researchers, lecturers, academic institutions, and LGBTQ+ organizations.

System Architecture



Technology Choices



Testing & Quality

~285 automated tests guard the full bilingual pipeline enforced on every commit

Unit — pytest · Jest · jest-axe — **~220** CASES

- Schemas, JWT auth & **RBAC** role hierarchy
- Hebrew NLP severity mapping & Unicode splitting
- React components + **WCAG/ARIA** via axe

Integration — pytest · vLLM — **End-to-end** API ↔ DB ↔ LLM

- Bilingual inference against live **vLLM**
- Docling PDF/DOCX parsing pipeline
- Private-mode **zero-retention** guarantees

Stress / Load — Locust — **Concurrent** USERS

- Sustained load on the **upload** endpoint
- Mixed Hebrew + English documents
- Latency & throughput under spawn ramp

CI/CD — GitHub Actions — **3** PIPELINES

- Lint + test **quality gate** on every PR
- Build & push images to **Azure ACR**
- Nightly vLLM integration run

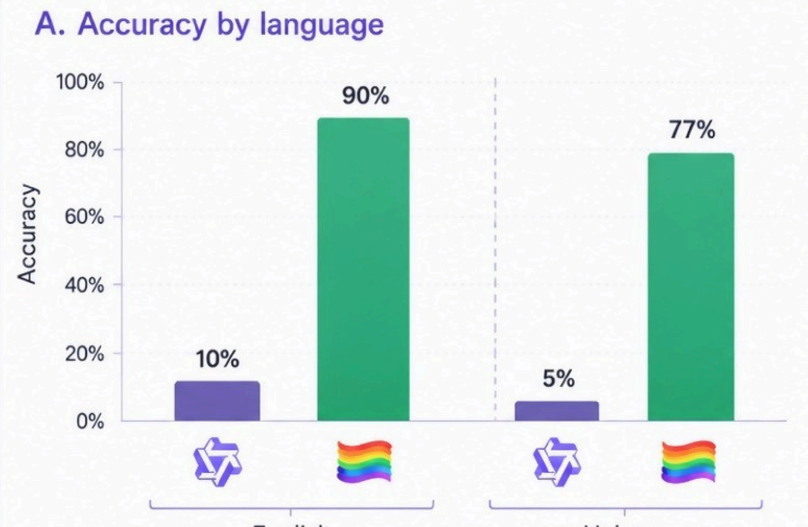
~285 tests | **32** test files | **3** frameworks (pytest · Jest · Locust) | **3** GitHub Actions pipelines

Evaluation Snapshot

Accuracy jumped to 83% overall — strong in both languages.

Inclusify decisively outperforms the base Qwen2.5-3B model.

A. Accuracy by language



B. Performance gains

Macro-F1
0.71
(base model = 0.12)
▲ 5.8× higher

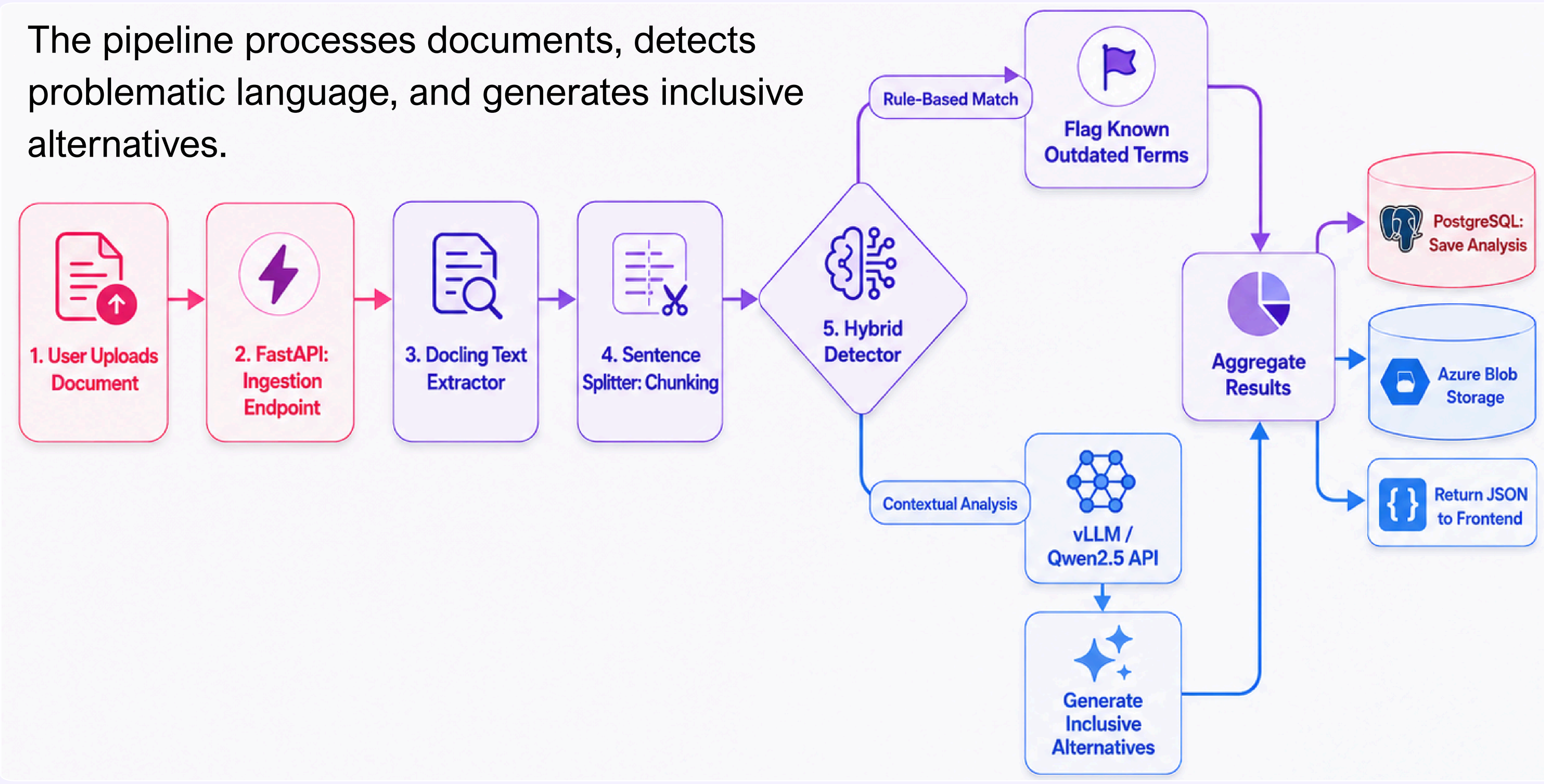
False positive
incorrect positive predictions
0.12
(base model = 1)
▼ from 100%

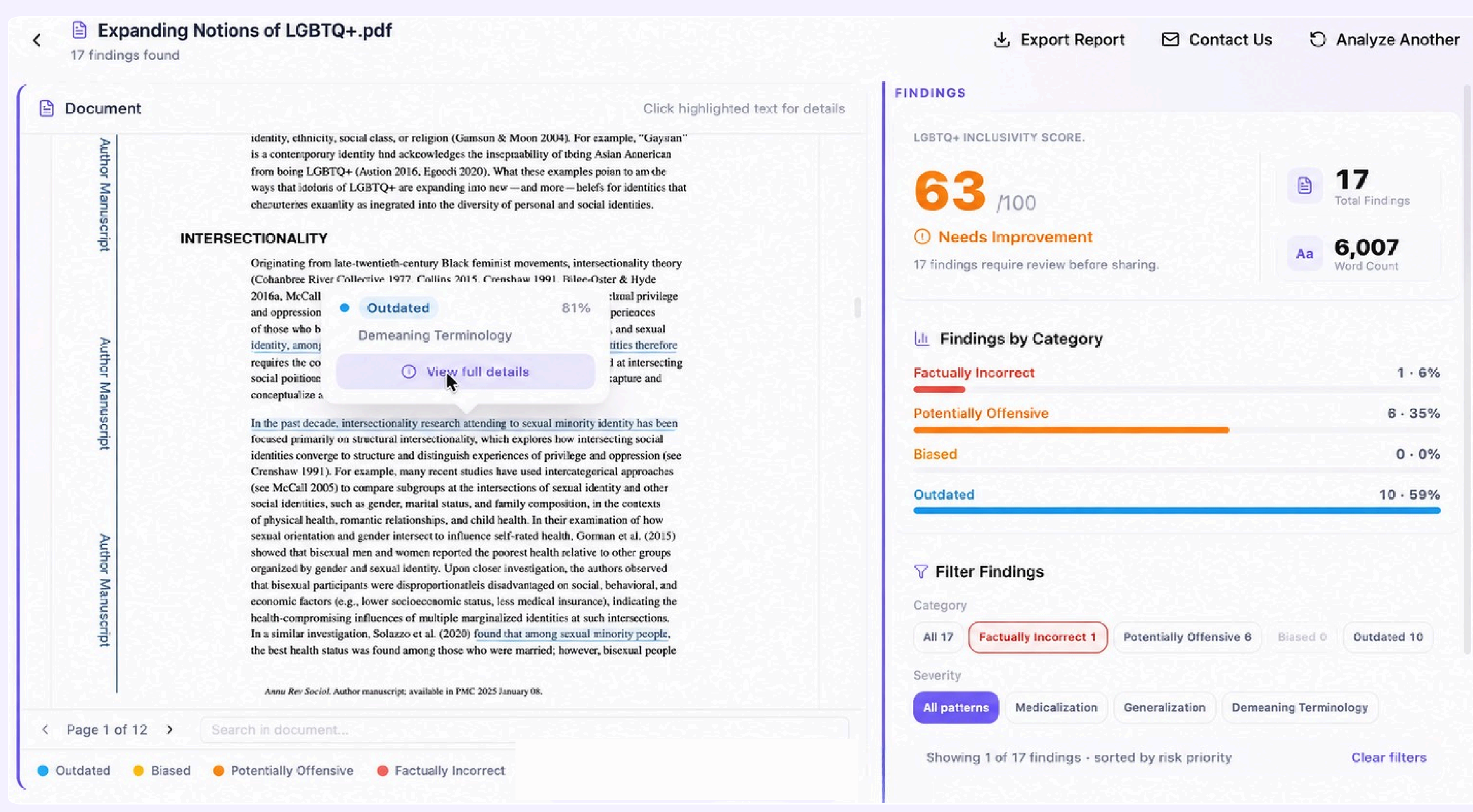
Trained on 14,092 examples (EN 6,380 • HE 6,302) | Test: 78 sentences (EN 39 • HE 39) | vs. base Qwen2.5-3B

How It Works

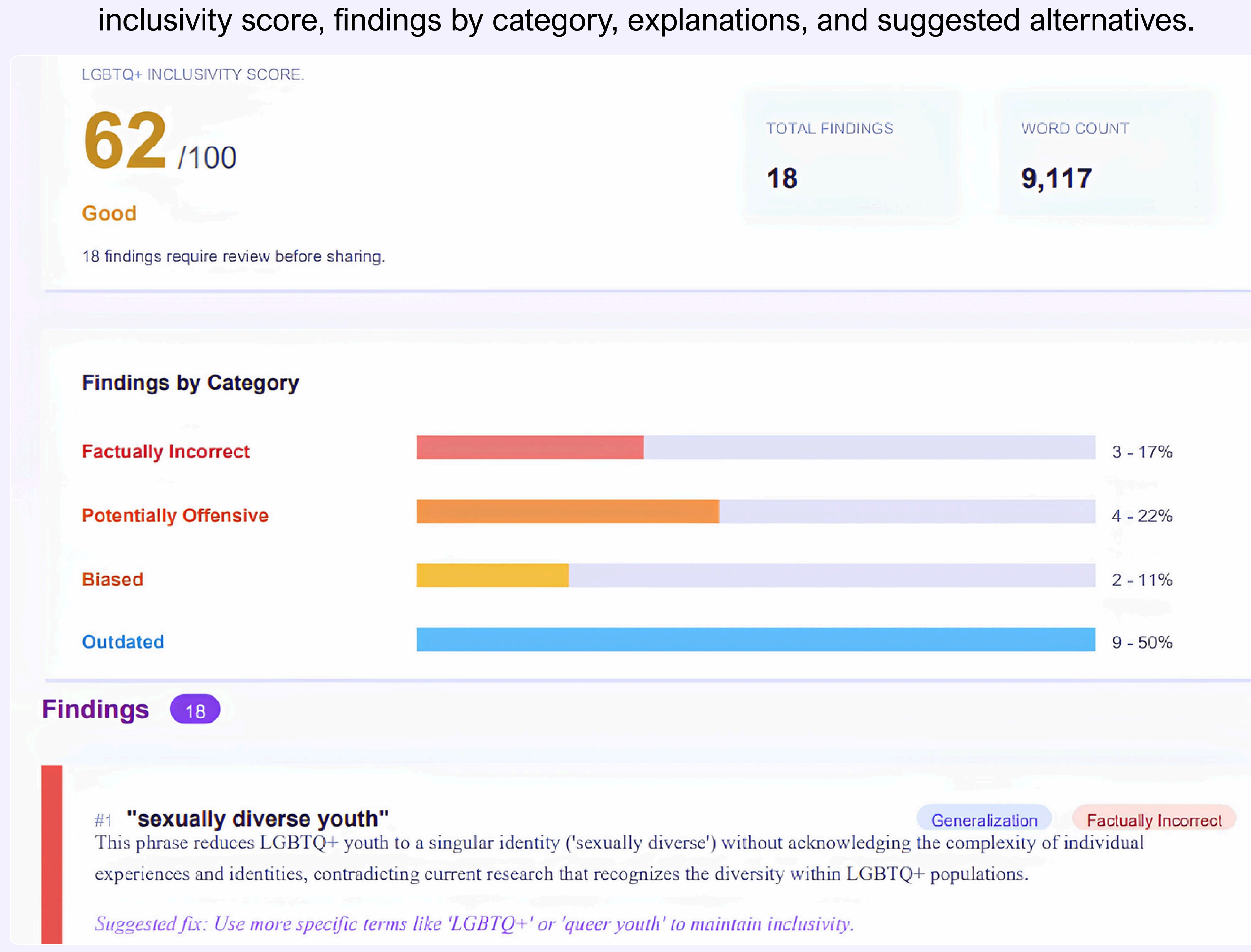
A bilingual, responsive and accessible interface designed for clear academic feedback in Hebrew and English.

The pipeline processes documents, detects problematic language, and generates inclusive alternatives.





In addition to the on-screen analysis, users can download a complete report with the inclusivity score, findings by category, explanations, and suggested alternatives.



Strengths

- Bilingual analysis in Hebrew and English
- Clear explanations and inclusive alternatives
- Privacy-focused and user-friendly design

Limitations

- May produce false-positive findings
- Hebrew text quality analysis needs improvement

Future Development

- Integrate an Audit Agent and RAG citations
- Improve Hebrew and outdated-term detection